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LOW WATER CONSUMPTION SHOWERHEAD

Abstract

A low water consumption showerhead may provide users with the experience of higher water pressure without necessarily requiring an increased supply of water. A first example showerhead may include nozzles having coaxial inner and outer chambers, each of which may provide a user with one or more different types of media. The inner chamber of the nozzle may be adjustable with respect to the outer chamber, which may allow a user to manipulate the relative rate of flow from both chambers. A second example showerhead may include a spiral channel along which one or more types of media may flow. Outlets may be positioned along the spiral channel such that a portion of the media within the spiral channel may exit the showerhead and be directed to a user. The width or area of the spiral channel may taper such that substantially consistent pressure may be provided at each outlet.

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Background/Summary

CROSS-REFERENCED TO RELATED APPLICATIONS [0001] The present application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/559,322, filed Feb. 29, 2024, entitled “LOW WATER CONSUMPTION SHOWERHEAD,” the entirety of which is hereby incorporated by reference.

BACKGROUND

[0002] Providing higher water pressure to showerheads traditionally involves supplying a higher volume of water. Therefore, high water pressure can result in higher utility payments, greater water usage and/or waste, and other undesirable outcomes. Accordingly, there is a need for a showerhead that provides increased water pressure without a higher volume of water.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0003] FIG. 1 is a bottom perspective view of an example showerhead.

[0004] FIG. 2 is a side cross-sectional view of an example showerhead.

[0005] FIG. 3 is a side cross-sectional view of an example nozzle.

[0006] FIG. 4 is a bottom plan view of an example nozzle.

[0007] FIG. 5 is a side cross-sectional view of an example adjustable nozzle in a first position.

[0008] FIG. 6 is a side cross-sectional view of an example adjustable nozzle in a second position.

[0009] FIG. 7 is a bottom plan view of an example adjustable nozzle.

[0010] FIG. 8 is a bottom perspective view of an example showerhead.

[0011] FIG. 9 is a top perspective view of an example showerhead.

[0012] FIG. 10 is a bottom cross-sectional view of an example showerhead.

[0013] FIG. 11 is a side cross-sectional view of an example heated conduit.

[0014] While the disclosure is susceptible to various modifications and alternative forms, a specific embodiment thereof is shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description presented herein are not intended to limit the disclosure to the particular embodiment disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

DETAILED DESCRIPTION

[0015] The invention will now be described with reference to the drawing figures, in which like reference numerals refer to like parts throughout. For purposes of clarity in illustrating the characteristics of the present invention, proportional relationships of the elements have not necessarily been maintained in the drawing figures.

[0016] Referring first to FIG. 1, an example showerhead 5 may have a body 10 defined by a faceplate 15 and a backplate 20, which may oppose the faceplate 15. When the showerhead 5 is in use by a user, the faceplate 15 may face in the direction of the user and the backplate 20 may face away from the user. A sidewall 25 may extend between outer boundaries 30 of the faceplate 15 and backplate 20. The faceplate 15 and backplate 20 may each be substantially circular in shape, and the sidewall 25 may thus have a generally circular profile. Although illustrated as circular in FIG. 1, the faceplate 15, backplate 20, and sidewall 25 may be provided in different shapes and sizes.

[0017] A primary inlet 35 may receive, be received by, or otherwise be coupled to an external

source of media (not shown) to allow a first type of media from the external source to enter the body **10** of the showerhead **5**. For example, the first type of media may be water, air, heated air, compressed air, hydrogen, ozone, essential oils, scented fluids, shampoo, conditioner, or any other fluid or gas. The external source may be, for example, a water pipe of a residential building. The primary inlet **35** may be coupled to or integrally formed with the backplate **20**. The primary inlet **35** may extend upwardly from the backplate **20**, or the primary inlet **35** may be positioned in any suitable manner.

[0018] One or more nozzles **40** may be in fluid communication with the primary inlet **35** such that the media that enters the showerhead **5** via the primary inlet **35** can exit the showerhead **5** via one or more of the nozzles **40**. The one or more nozzles **40** may be coupled to and extend downwardly from the faceplate **15**, and the one or more nozzles **40** may be provided in the form of openings positioned on the faceplate **15** or in other forms now known or hereafter developed. In the example of FIG. **1**, the showerhead **5** includes six nozzles **40** disposed proximate to the outer boundary **30** of the faceplate **15** and evenly spaced radially apart from one another. However, the showerhead **5** may include any number of nozzles **40** which may be positioned in any suitable arrangement.

[0019] Turning to FIG. **2**, the showerhead **5** may include one or more secondary inlets (not shown) in addition to or in place of the primary inlet **35**. The one or more secondary inlets may receive, be received by, or otherwise be coupled to an external source of media (not shown) to allow a second type of media from the external source to enter the showerhead **5**. For example, the second type of media may be water, air, heated air, compressed air, hydrogen, ozone, essential oils, scented fluids, shampoo, conditioner, or any other fluid or gas. The external source that feeds the one or more secondary inlets may be the same external source or a different external source than the external source that feeds the primary inlet **35**. The first type of media that enters the body **10** via the primary inlet **35** may be the same type of media or a different type of media than the second type of media that enters the body **10** via the one or more secondary inlets. As one example, water may enter the body **10** of the showerhead **5** via the primary inlet **35** and air may enter the body **10** via the one or more secondary inlets. As another example, water may enter the body **10** via both the primary inlet **35** and the one or more secondary inlets. Alternatively, the showerhead **5** may have any other suitable configuration.

[0020] A partition **45** may be positioned within the showerhead **5** to separate the first and second type of media within the body **10**. The partition **45** may be disposed within the body **10**, and at least a portion of the partition **45** may extend into and occupy at least a portion of at least one nozzle **40**. The partition **45** may divide at least one nozzle **40** into two distinct portions—a first chamber **50**, which may be an inner chamber, and a second chamber **55**, which may be an outer chamber—that are not in fluid communication with one another. The second chamber **55** may at least partially surround the first chamber **50**. Thus, the partition **45** may prevent the first and second type of media that enter the showerhead **5** from mixing until they exit the showerhead **5** via one or more nozzles **40**. The partition **45** may at least partially extend into each nozzle **40** such that each nozzle **40** is divided into the first chamber **50** and the second chamber **55**.

[0021] Turning to FIG. **3**, a portion of the partition **45** that extends into the nozzle **40** may form a first housing **60**. The nozzle **40** may include the first chamber **50**. The first chamber **50** may be defined by the first housing **60**. The nozzle **40** may include the second chamber **55**. The second chamber **55** may be defined by the space between the first housing **60** and a second housing **65**. The overall shape of the nozzle **40**, which may be defined by the second housing **65**, may be substantially conical or frustoconical or may have another shape. The shape of the first chamber **50**, which may be defined by the first housing **60**, may be substantially cylindrical or may have another shape.

[0022] The first and second housings **60**, **65** may be positioned along a central longitudinal axis A of the nozzle **40**. The first housing **60** and the second housing **65** may be angled with respect to the axis A, or may be substantially parallel to the axis A. For example, in the configuration of FIG. **3**,

the distance between the second housing **65** and the axis A (which may be measured in a direction substantially perpendicular to the axis A) may be greater at a first end **70** of the nozzle **40** than it is at a second end **75** of the nozzle **40**. As such, a width W of the second chamber **55**, which may be associated with the distance between the first housing **60** and the second housing **65**, may taper from a first width at the first end **70** to a second width at the second end **75**. The first width of the second chamber **55** at the first end **70** may be greater than the second width of the second chamber **55** at the second end **75**. Alternatively, the first housing **60** and the second housing **65** may each be oriented differently with respect to the axis A. For example, the first housing **60** and second housing **65** may each be oriented substantially parallel to the axis A, or may each be oriented at a same or different angle with respect to the axis A. The first housing **60** and second housing **65** may be oriented in any suitable configuration which may be selected to achieve a particular result.

[0023] The first chamber **50** may include a first inlet **80**. The first inlet **80** may form an opening between the first chamber **50** and the body **10** of the showerhead **5** at the first end **70** of the nozzle **40**. The first inlet **80** may allow media that enters the body **10** to travel to the first chamber **50**. The first chamber **50** may include a first outlet **85**. The first outlet **85** may form an opening of the first chamber **50** to the external environment to allow media that enters the first chamber **50** to exit the showerhead **5** at the second end **75** of the nozzle **40**.

[0024] Likewise, the second chamber **55** may include a second inlet **90**. The second inlet **90** may form an opening between the second chamber **55** and the body **10** of the showerhead **5** at the first end **70** of the nozzle **40**. The second inlet **90** may allow media that enters the body **10** to travel to the second chamber **55**. The second chamber **55** may include a second outlet **95**. The second outlet **95** may form an opening of the second chamber **55** to the external environment to allow media that enters the second chamber **55** to exit the showerhead **5** at the second end **75** of the nozzle **40**.

[0025] The first chamber **50** and second chamber **55** may receive media from one or more external sources via the first inlet **80** and second inlet **90**, respectively. The first chamber **50** and second chamber **55** may direct such media out of the showerhead **5** and toward a user via the first outlet **85** and second outlet **95**, respectively. The first outlet **85** and second outlet **95** may expel different types of media or the same type of media. For example, the first outlet **85** may expel media that enters the body **10** via the one or more secondary inlets, whereas the second outlet **95** may expel media that enters the body **10** via the primary inlet **35**. Alternatively, the first outlet **85** and second outlet **95** may each expel media that enters the body **10** via the primary inlet **35** or may be configured in any other suitable arrangement. When the showerhead **5** is in use, media may be provided to a user via only the first outlet **85**, only the second outlet **95**, or both the first outlet **85** and second outlet **95** simultaneously.

[0026] As a non-limiting example, air may flow from an external source into the body **10** of the showerhead **5** via the one or more secondary inlets. Such air may enter the first chamber **50** of the nozzle **40** via the first inlet **80** and may exit the nozzle **40** via the first outlet **85**. At the same time, water may flow from an external source into the body **10** of the showerhead **5** via the primary inlet **35**. Such water may enter the second chamber **55** of the nozzle **40** via the second inlet **90** and may exit the nozzle **40** via the second outlet **95**. In this example, the first housing **60** may keep the air and water separated within the nozzle **40**, but the air and water may mix or blend to form a unitary stream after exiting the nozzle **40**. Such mixing or blending may occur before the media comes into contact with a user. In this example, the air may increase the force of the stream flowing from each nozzle **40** such that a user may perceive the sensation of a stream with higher water pressure without increasing the volume of water supplied to the showerhead **5**. Thus, the addition of air or another non-water media to the stream provided by the nozzle **40** may enable a user to reduce water usage without sacrificing the quality of the experience of using the showerhead **5**.

[0027] As illustrated in FIG. 4—a bottom plan view of the nozzle **40**—the first outlet **85** may be substantially circular in shape and the second outlet **95** may be substantially annular in shape. As such, both the first housing **60** and the second housing **65** may be provided in the form of an

annular cylinder. Thus, in the illustrated example, the first outlet **85**, first housing **60**, second outlet **95**, and second housing **65** may all have substantially circular profiles when viewed from below. However, other shapes are envisioned for the first outlet **85**, first housing **60**, second outlet **95**, and second housing **65**, and each of the foregoing may be any suitable shape or size.

[0028] Turning to FIG. 5, a nozzle **100** may be constructed so that a portion of the nozzle **100** may be adjustable by translation in a linear direction such that the flow of a first and second type of media from the nozzle **100** may be varied. The nozzle **100** may be constructed similarly to the nozzle **40** as described above. The nozzle **100** may include a first chamber **105** and a second chamber **115**. The first chamber **105** may be adjustable and may be defined by a first housing **110**. The second chamber **115** may be defined by a second housing **120**. The first chamber **105** may be an inner chamber and the second chamber **115** may be an outer chamber.

[0029] The first chamber **105** of the nozzle **100** may be movable between a first position and a second position. The direction of movement of the first chamber **105**, which may follow a path along arrow M, may be substantially parallel with an axis A. Thus, the first chamber **105** may be adjustable relative to the second chamber **115** of the nozzle **100**. When the first chamber **105** is in the first position, as one example and as illustrated in FIG. 5, a first end **125** of the first housing **110** may be proximate to or substantially coplanar with a first end **130** of the second housing **120**, while a second end **135** of the first housing **110** may extend beyond a second end **140** of the second housing **120**.

[0030] A first outlet **145** may form an opening of the first chamber **105** to the external environment, and a second outlet **150** may form an opening of the second chamber **115** to the external environment. A width W1 of the first outlet **145** may be fixed, while a width W2 of the second outlet **150** may vary depending on the position of the first chamber **105** with respect to the second housing **120**. As the first chamber **105** moves between the first and second positions, the distance between the first housing **110** and the second housing **120** may change, thereby changing the width W2 of the second outlet **150**. Therefore, a user may manipulate the volume of the second chamber **115** and the width W2 of the second outlet **150** by adjusting the position of the first chamber **105** with respect to the second housing **120**. By doing so, a user may manipulate the rate of flow of media from the second outlet **150**, and as a result, the user may manipulate the ratio of the rate of flow of media from the second outlet **150** relative to the rate of flow of media from the first outlet **145**.

[0031] As illustrated in FIG. 6, the first chamber **105**, defined by the first housing **110**, of the nozzle **100** may occupy a second position with respect to the second housing **120**. With the first chamber **105** in the second position, the second end **135** of the first housing **110** may be proximate to or substantially coplanar with the second end **140** of the second housing **120**, while the first end **125** of the first housing **110** may extend beyond the first end **130** of the second housing **120**. In this configuration, the first outlet **145** may have a width W3 and the second outlet **150** may have a width W4. The width W3 may be the same as the width W1 of FIG. 5, whereas the width W4 may be greater than the width W2 of FIG. 5. Accordingly, the volume of the second chamber **115** when the first chamber **105** is in the second position may be greater than the volume of the second chamber **115** when the first chamber **105** is in the first position. The second chamber **115** may be adjustable relative to the first chamber **105** in the same manner as described herein with reference to the first chamber **105** being adjustable relative to the second chamber **115**.

[0032] Turning to the bottom plan view of the nozzle **100** provided in FIG. 7, the first outlet **145** may be substantially circular in shape and the second outlet **150** may be substantially annular in shape. As such, both the first housing **110** and the second housing **120** may be provided in the form of an annular cylinder. Thus, in the illustrated example, the first outlet **145**, first housing **110**, second outlet **150**, and second housing **120** may all have substantially circular profiles when viewed from below. However, other shapes are envisioned for the first outlet **145**, first housing **110**, second outlet **150**, and second housing **120**, and each of the foregoing may be any suitable shape or

size.

[0033] The configuration illustrated in FIG. 7 may correspond to the first position or the second position of the first chamber **105**, as illustrated in FIG. 5 and FIG. 6, respectively. The width **W5** of the first outlet **145** may be fixed, whereas the width **W6** of the second outlet **150** may vary according to the position of the first chamber **105** with respect to the second housing **120**. As one example, the widths **W5**, **W6** may correspond to the widths **W1**, **W2** of FIG. 5, respectively. As another example, the widths **W5**, **W6** may correspond to the widths **W3**, **W4** of FIG. 6, respectively. Alternatively, other widths may correspond to the widths **W5**, **W6** depending on the positioning of the first chamber **105**. As the first chamber **105** moves between the first and second position, the width **W6** and the ratio between the rate of flow of media from the first outlet **145** and the rate of flow of media from the second outlet **150** may vary.

[0034] Turning to FIG. 8, a second example showerhead **155** may have a body **160** defined by a faceplate **165**, which may have a center point **C**, and a backplate **170** which may oppose the faceplate **165**. A sidewall **175** may extend between outer boundaries **180** of the faceplate **165** and the backplate **170**. The faceplate **165** and backplate **170** may each be substantially circular in shape, and the sidewall **175** may thus have a generally circular profile. Although illustrated as circular in FIG. 1, the faceplate **165**, backplate **170**, and sidewall **175** may be provided in different shapes. The showerhead **155** may be constructed such that, when the showerhead **155** is in use by a user, the faceplate **165** faces in the direction of the user while the backplate **170** faces away from the user.

[0035] One or more inlets **185** may receive, be received by, or otherwise be coupled to an external source of media (not shown) and allow media from the external source to enter the body **160** of the showerhead **155**. For example, the media may be water, air, heated air, compressed air, hydrogen, ozone, essential oils, scented fluids, shampoo, conditioner, or any other fluid or gas. The external source may be, for example, a water pipe of a residential building. The one or more inlets **185** may be coupled to or formed integrally with the backplate **170**. The one or more inlets **185** may extend upwardly from the backplate **170**, or the one or more inlets **185** may be positioned in any suitable manner.

[0036] The showerhead **155** may include a plurality of outlets **190** arranged on the faceplate **165**. Each of the plurality of outlets **190** may be in fluid communication with one or more inlets **185** such that the one or more inlets **185** may provide media to the plurality of outlets **190**. Each of the plurality of outlets **190** may form an opening in the faceplate **165** through which media that enters the body **160** of the showerhead **155** via one or more inlets **185** may exit the showerhead **155** to the external environment. When the showerhead **155** is in use, upon exiting the showerhead **155**, media may be directed toward a user. In the example of FIG. 8, the showerhead **155** may include thirty-six outlets **190** which may be arranged in a pattern of radial spokes extending between the center point **C** and various points along the outer boundary **180** of the faceplate **165**. Alternatively, the showerhead **155** may include any number of outlets **190**, and the outlets **190** may be positioned in any suitable arrangement.

[0037] Turning to FIG. 9, the showerhead **155** may include a first inlet **185** positioned proximate to the outer boundary **180** of the backplate **170**. The first inlet **185** may be substantially L-shaped and may include at least a first opening **195** and a second opening **200**. In the example of FIG. 9, the first opening **195** and the second opening **200** may be oriented substantially perpendicular to one another. However, the first and second openings **195**, **200** may be oriented at any other suitable angle with respect to one another. Alternatively, the first inlet **185** may be any other suitable shape and may only include the first opening **195**.

[0038] The showerhead **155** may also include a second inlet **205**. The second inlet **205** may allow for media from an external source to enter the body **160** of the showerhead **155**. For example, the media entering via the second inlet **205** may be water, air, heated air, compressed air, hydrogen, ozone, essential oils, scented fluids, shampoo, conditioner, or any other fluid or gas. The external source that feeds the second inlet **205** may be the same external source or a different external

source than the external source that feeds the first inlet **185**. The second inlet **205** may be positioned at a distance from the outer boundary **180** of the backplate **170**. For example, the second inlet **205** may be substantially aligned with the center point C of the faceplate **165**.

[0039] Various configurations of the first and second inlets **185**, **205** are envisioned for providing one or more types of media to the body **160** of the showerhead **155**. For example, the first inlet **185** may allow two different types of media to enter the body **160** and media may be prevented from entering the body **160** via the second inlet **205**. By way of example, the first opening **195** may receive, be received by, or otherwise be coupled to an external source of water, and the second opening **200** may receive, be received by, or otherwise be coupled to an external source of compressed air. In this configuration, media may enter the body **160** exclusively through the first inlet **185** and the second inlet **205** may be blocked to prevent additional media from unintentionally entering the body **160**.

[0040] Alternatively, the first inlet **185** may allow a first type of media to enter the body **160**, and the second inlet **205** may allow a second type of media to enter the body **160**. The first and second types of media entering the body **160** via the first inlet **185** and second inlet **205**, respectively, may be the same type of media, or they may be different. By way of example, in this configuration, the first opening **195** of the first inlet **185** may receive, be received by, or otherwise be coupled to an external source of water. The second opening **200** may be blocked to prevent additional media from unintentionally entering the body **160**. At the same time, the second inlet **205** may receive, be received by, or otherwise be coupled to an external source of compressed air. In this configuration, the water entering the body **160** via the first inlet **185** and the compressed air entering the body **160** via the second inlet **205** may produce a resonant frequency or standing wave that may result in a “pulsing” behavior of the media that ultimately exits the body **160** and is provided to a user.

[0041] In yet other alternatives, the first inlet **185** and second inlet **205** may be provided in any other suitable configuration and may allow for any media to enter the body **160**. For example, the first and second openings **195**, **200** of the first inlet **185** may both be blocked, and media may be provided to the body **160** exclusively via the second inlet **205**. In any configuration in which air enters the body **160** via either the first inlet **185** or the second inlet **205**, the air may be compressed air. Configurations that provide multiple types of media to a user may cause the user to perceive the sensation of a stream with higher water pressure without increasing the volume of water supplied to the showerhead **155**.

[0042] Turning to FIG. **10**, the body **160** may house a spiral channel **210** (which may be defined by a sidewall **215**). The spiral channel **210** may define a pathway along which media may flow through the showerhead **155**. The spiral channel **210** may have a starting end **220** proximate to the outer boundaries **180** and a terminal end **225** proximate to the center point C. Thus, media from one or more inlets **185**, the second inlet **205**, or combinations thereof may enter the spiral channel **210** at the starting end **220** or proximate to the center point C. The media may travel along the spiral channel **210** as the spiral channel **210** revolves around and approaches closer to the center point C until the media reaches the terminal end **225**. The outlets **190** may be positioned on the faceplate **165** to align with the spiral channel **210** such that each outlet **190** forms an opening of the spiral channel **210** to the external environment. As a result, gravity and/or pressure within the spiral channel **210** may cause a portion of the media flowing within the spiral channel **210** to exit via one or more of the outlets **190** before reaching the terminal end **225**. When more than one type of media enters the body **160** of the showerhead **155**, the multiple types of media may mix within the spiral channel **210** before exiting the body **160** and being directed toward a user.

[0043] The spiral channel **210** may have a tapered geometry, which may influence the pressure at which media is provided at each outlet **190**. For example, the spiral channel **210** may have a starting width **W7**, which may be located at the starting end **220**, and may have a terminal width **W8**, which may be located at the terminal end **225**. The starting width **W7** may be greater than the terminal width **W8**, and the width of the spiral channel **210** may taper from the starting width **W7**

to the terminal width **W8** along the length of the spiral channel **210**.

[0044] Likewise, the spiral channel **210** may have a cross-sectional area, which may be defined by a cross-section of the spiral channel **210** taken in a direction that is substantially perpendicular to the faceplate **165** at any point along the length of the spiral channel **210**. The cross-sectional area of the spiral channel **210** may decrease as the width of the spiral channel tapers from the starting width **W7** to the terminal width **W8** along the length of the spiral channel **210**. For example, the spiral channel **210** may have a first cross-sectional area at the starting end **220** and a second cross-sectional area at the terminal end **225**, and the first cross-sectional area may be greater than the second cross-sectional area.

[0045] The tapered geometry of the spiral channel **210** may provide media at a substantially consistent pressure at each outlet **190**. For example, narrowing of the spiral channel **210** as water travels from the starting end **220** to the terminal end **225** may increase the localized pressure within the spiral channel **210**. The increase in localized pressure due to the geometry of the spiral channel **210** may approximately compensate for the loss in pressure that may result as at least a portion of the volume of media within the spiral channel **210** is lost to the external environment via one or more outlets **190**.

[0046] The rate of tapering of the width and the cross-sectional area of the spiral channel **210** may or may not be continuous. For example, the rate of tapering from the starting end **220** to the terminal end **225** may be constant. Alternatively, the rate of tapering may be discontinuous and the width and cross-sectional area of the spiral channel **210** may change in “steps.” In yet other alternatives, the rate of tapering may increase or decrease along the length of the spiral channel **210**. The specific parameters defining the tapered geometry of the spiral channel **210** may be manipulated or optimized via computer modeling or testing to influence the pressure at which media is provided at each outlet **190**. For example, computer modelling may be used to ensure that media is provided at the same pressure at each outlet **190**.

[0047] In the example of FIG. **10**, the width and cross-sectional area of the spiral channel **210** at the starting end **220** may be greater than the width and cross-sectional area of the spiral channel at the terminal end **225**. Alternatively, the inverse may be true. For example, the width and cross-sectional area of the spiral channel **210** at the terminal end **225** may be greater than the width and cross-sectional area of the spiral channel **210** at the starting end **220**. Additionally, in the example of FIG. **10**, the spiral channel **210** is provided in the form of a spiral having a regular or symmetrical shape. However, the spiral channel **210** may be alternatively provided in the form of an irregular spiral, a serpentine shape, an S-shaped pattern, or any other suitable shape or configuration.

[0048] Turning to FIG. **11**, a heated conduit **230** may be included in the showerhead **5**, the showerhead **155**, or other examples of the present invention. The heated conduit **230** may heat media within a showerhead before it is directed toward a user. For example, when the showerhead **5** provides a mixture of water and heated air to a user, the heated conduit **230** may provide the means for warming the air into heated air before directing the heated air toward a user. The heated conduit **230** may facilitate humidification of the heated air by imparting moisture to the air during heating. This humidification may, for example, increase the heat content of the heated air or improve the perceived comfort of a user when being impacted by humidified heated air as opposed to dry heated air.

[0049] The heated conduit **230** may be used to heat water or another media, and the heated conduit **230** may heat more than one type of media simultaneously. Further, the heated conduit **230** may heat a mixture containing multiple types of media. For example, the heated conduit **230** may apply heat to a mixture of air and water, rather than heating the air and water separately before they combine to form a mixture. In this way, the heated conduit may impart heat to the mixture with a more even distribution than would be possible if multiple types of media are heated separately before combining to form a mixture. For example, the heated conduit **230** may heat a mixture of multiple types of media without imparting substantially more heat to one portion of the mixture

than to another. The improved heat distribution provided by the heated conduit **230** may provide a heated mixture of media to a user without producing a portion of the mixture that may scald or otherwise harm the user. The heated conduit **230** may be substantially similar in structure and function to the conduit assembly described in U.S. Pat. No. 11,331,246 assigned to Kohler Co., the entirety of which is hereby incorporated by reference.

[0050] A blow dryer may be integrated into the showerhead **5**, the showerhead **155**, or other examples of the present invention. The blow dryer may be utilized to assist in drying a user after showering. The blow dryer may output heated air from the showerhead **5** or showerhead **155**, which may be directed through the outlets onto the user.

[0051] As is evident from the foregoing description, certain aspects of the present invention are not limited by the particular details of the examples illustrated herein, and it is therefore contemplated that other modifications, applications, variations, or equivalents thereof, will occur to those skilled in the art. Many such changes, modifications, variations, and other uses and applications of the present constructions will, however, become apparent to those skilled in the art after considering the specification and the accompanying drawings. In addition, unless mention was made above to the contrary, it should be noted that all of the accompanying drawings are not to scale. All such changes, modifications, variations, and other uses and applications which do not depart from the spirit and scope of the present inventions are deemed to be covered by the inventions which are limited only by the claims which follow.

Claims

1. A showerhead comprising: a plurality of nozzles, wherein each of the plurality of nozzles comprises: a first chamber defined by a first housing; and a second chamber defined by the first housing and a second housing; at least two inlets, wherein a first inlet of the at least two inlets provides a first media to the first chamber, and wherein a second inlet of the at least two inlets provides a second media to the second chamber; wherein the first media and the second media mix in air after exiting a first outlet of the first chamber and a second outlet of the second chamber, respectively.
2. The showerhead of claim 1, further comprising a partition that divides each of the plurality of nozzles into the first chamber and the second chamber.
3. The showerhead of claim 1, wherein the first chamber is adjustable relative to the second chamber between a first position and a second position.
4. The showerhead of claim 3, wherein when the first chamber is in the first position, the second chamber has a first width, and wherein when the first chamber is in the second position, the second chamber has a second width.
5. The showerhead of claim 4, wherein the first width is less than the second width.
6. The showerhead of claim 3, wherein when the first chamber is in the first position, the second chamber has a first cross-sectional area, and wherein when the first chamber is in the second position, the second chamber has a second cross-sectional area.
7. The showerhead of claim 6, wherein the first cross-sectional area is less than the second cross-sectional area.
8. The showerhead of claim 1, wherein the second chamber is adjustable relative to the first chamber.
9. A showerhead comprising: a substantially spiral-shaped channel, wherein the channel comprises a first end and a second end, and wherein a first width of the channel proximate to the first end is greater than a second width of the channel proximate to the second end; at least one inlet for providing a media to the channel; and a plurality of nozzles aligned along the channel; wherein a first pressure of media exiting a first nozzle of the plurality of nozzles is substantially equal to a second pressure of media exiting a second nozzle of the plurality of nozzles.

10. The showerhead of claim 9, wherein each of the plurality of nozzles is arranged in a pattern of radial spokes extending between a center point of a body of the showerhead and an outer edge of the body.

11. The showerhead of claim 9, wherein the at least one inlet comprises at least one of: a first inlet coupled to a first external source of a first media; and a second inlet coupled to a second external source of a second media; wherein the first inlet and the second inlet are each in fluid communication with the channel.

12. The showerhead of claim 11, wherein the first media and the second media mix within the channel before being dispensed by the plurality of nozzles.

13. The showerhead of claim 11, wherein the first media is water and the second media is air.

14. The showerhead of claim 11, wherein the first media is air and the second media is water.

15. The showerhead of claim 9, further comprising a heated conduit which heats the media entering the channel.

16. A showerhead comprising: a plurality of nozzles, wherein each of the plurality of nozzles comprises: a first chamber; and a second chamber; at least two inlets, wherein a first inlet of the at least two inlets provides a first media to the first chamber, and wherein a second inlet of the at least two inlets provides a second media to the second chamber; wherein the first chamber is adjustable relative to the second chamber between a first position and a second position.

17. The showerhead of claim 16, wherein the first media and the second media mix in air after exiting a first outlet of the first chamber and a second outlet of the second chamber, respectively.

18. The showerhead of claim 16, wherein when the first chamber is in the first position, the second chamber has a first width, and wherein when the first chamber is in the second position, the second chamber has a second width.

19. The showerhead of claim 18, wherein the first width is less than the second width.

20. The showerhead of claim 16, wherein when the first chamber is in the first position, the second chamber has a first cross-sectional area, and wherein when the first chamber is in the second position, the second chamber has a second cross-sectional area, and wherein the first cross-sectional area is less than the second cross-sectional area.
